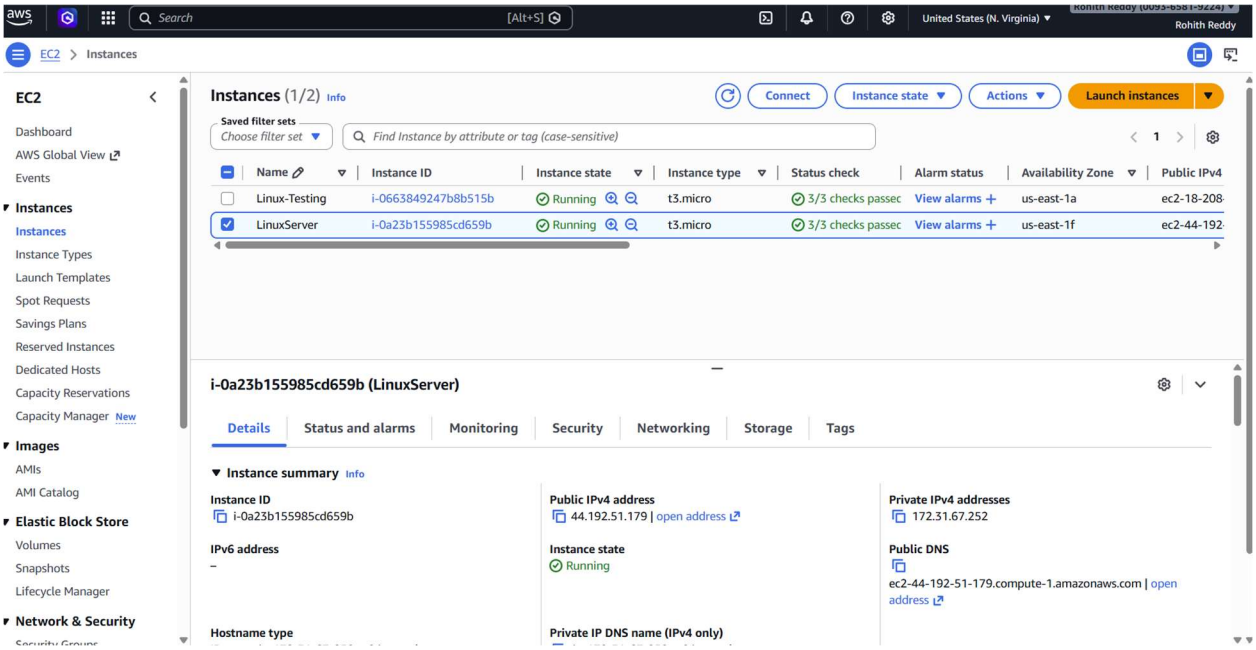
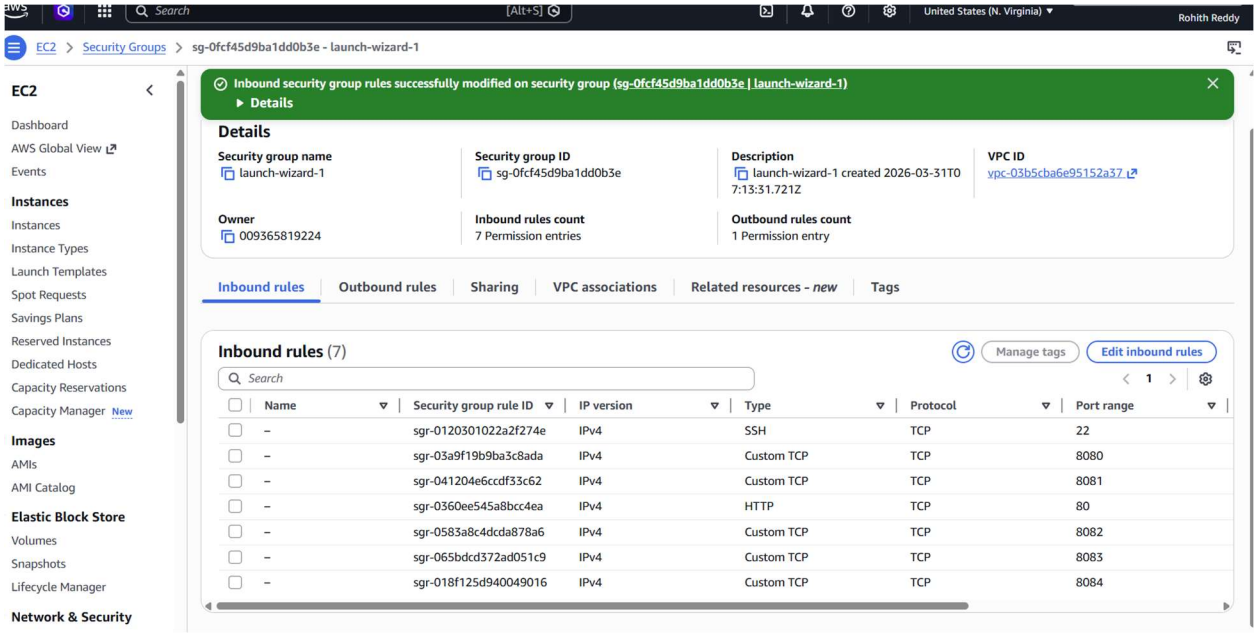
Setting up a single EC2 instance using Amazon Linux



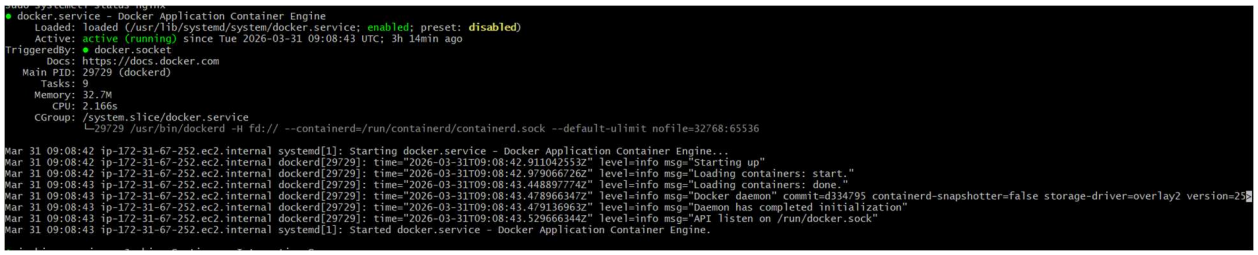
and configure the following services:

Service	**Port**	**Why**
SSH	22	For Remote access
Jenkins	8081	For CI/CD
Apache	8082	For Web server
Nginx	8083	For Web server
Tomcat	8084	For Java apps

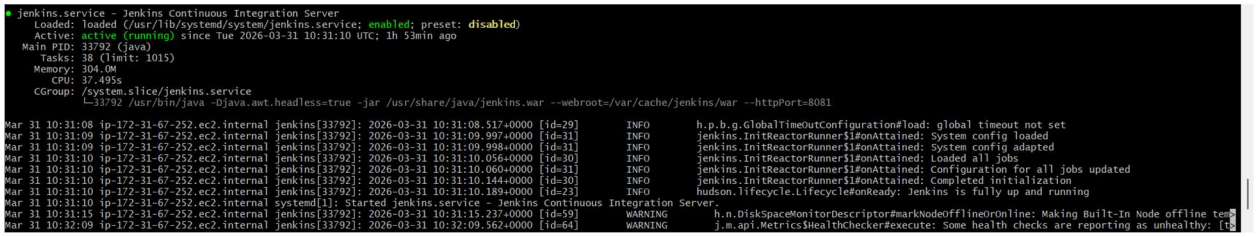
Assigned **unique ports** For each services

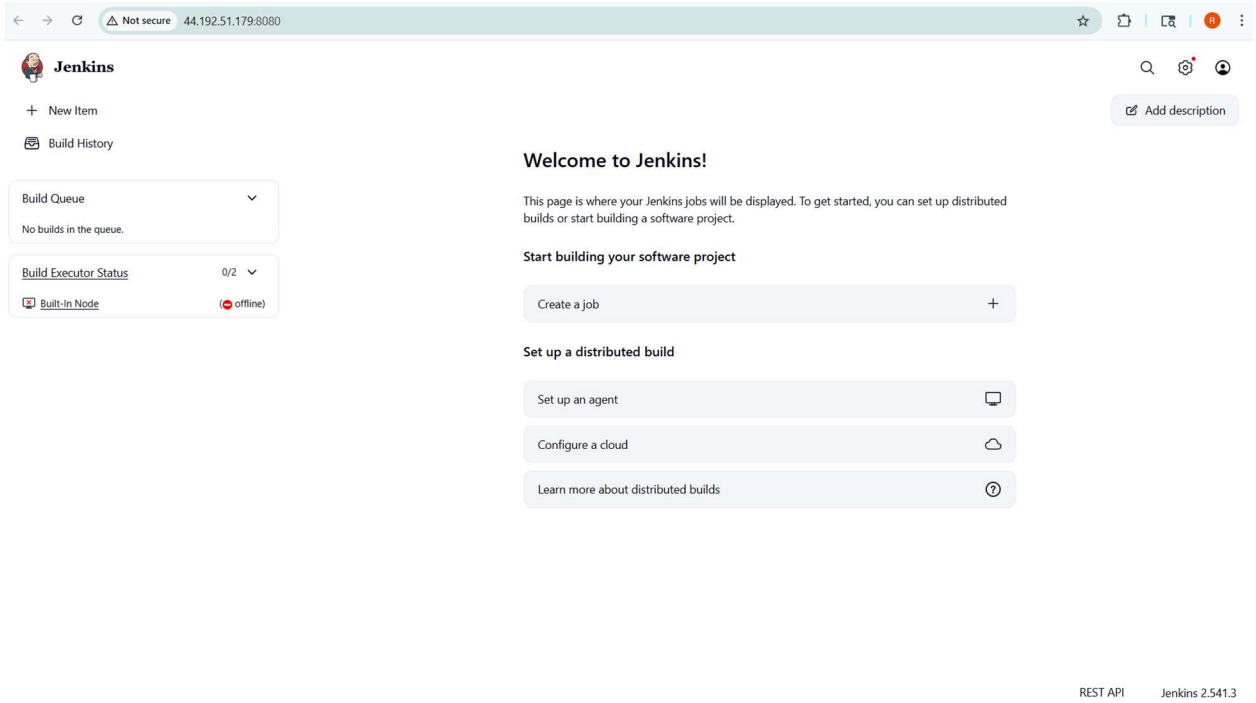


- Installed Docker and successfully running



- Installed Jenkins and successfully running



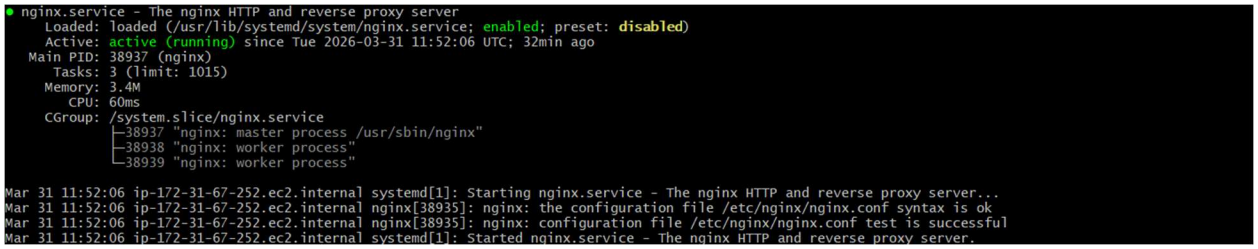


- Installed Apache (httpd) and successfully running

```
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Tue 2026-03-31 11:43:00 UTC; 41min ago
       Docs: man:httpd.service(8)
   Main PID: 37650 (httpd)
     Status: "Total requests: 3; Idle/Busy workers 100/0;Requests/sec: 0.0012; Bytes served/sec:   1 B/sec"
      Tasks: 230 (limit: 1015)
     Memory: 17.3M
        CPU: 2.547s
     CGroup: /system.slice/httpd.service
             ├─37650 /usr/sbin/httpd -DFOREGROUND
             ├─37651 /usr/sbin/httpd -DFOREGROUND
             ├─37652 /usr/sbin/httpd -DFOREGROUND
             ├─37653 /usr/sbin/httpd -DFOREGROUND
             ├─37654 /usr/sbin/httpd -DFOREGROUND
             └─37978 /usr/sbin/httpd -DFOREGROUND

Mar 31 11:43:00 ip-172-31-67-252.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Mar 31 11:43:00 ip-172-31-67-252.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Mar 31 11:43:00 ip-172-31-67-252.ec2.internal httpd[37650]: Server configured, listening on: port 8082
```

- Installed Nginx and successfully running

- 
```
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
     Active: active (running) since Tue 2026-03-31 11:52:06 UTC; 32min ago
   Main PID: 38937 (nginx)
      Tasks: 3 (limit: 1015)
     Memory: 3.4M
        CPU: 60ms
     CGroup: /system.slice/nginx.service
             ├─38937 "nginx: master process /usr/sbin/nginx"
             ├─38938 "nginx: worker process"
             └─38939 "nginx: worker process"

Mar 31 11:52:06 ip-172-31-67-252.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Mar 31 11:52:06 ip-172-31-67-252.ec2.internal nginx[38935]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Mar 31 11:52:06 ip-172-31-67-252.ec2.internal nginx[38935]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Mar 31 11:52:06 ip-172-31-67-252.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
```

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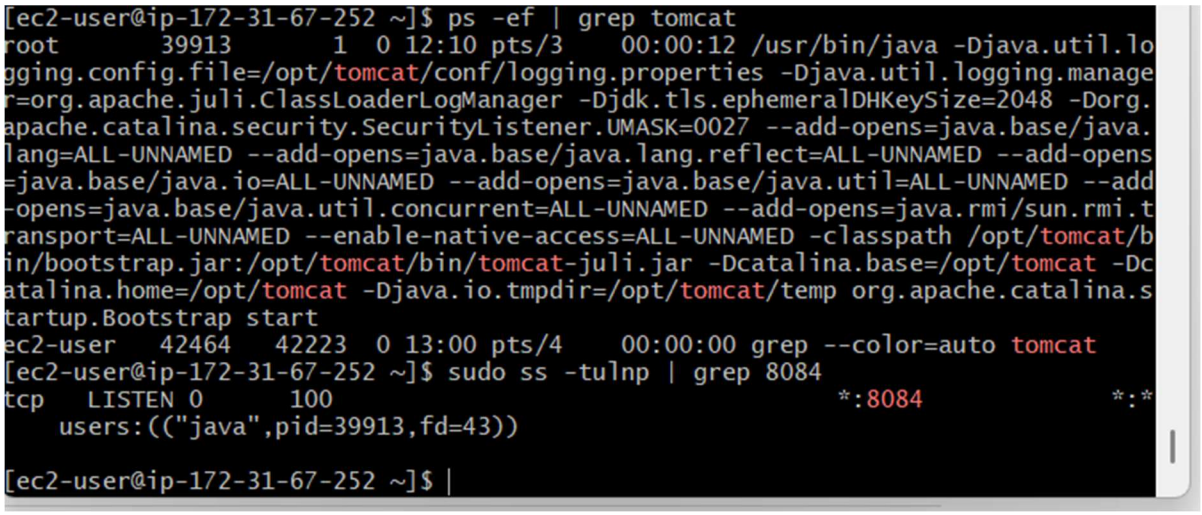
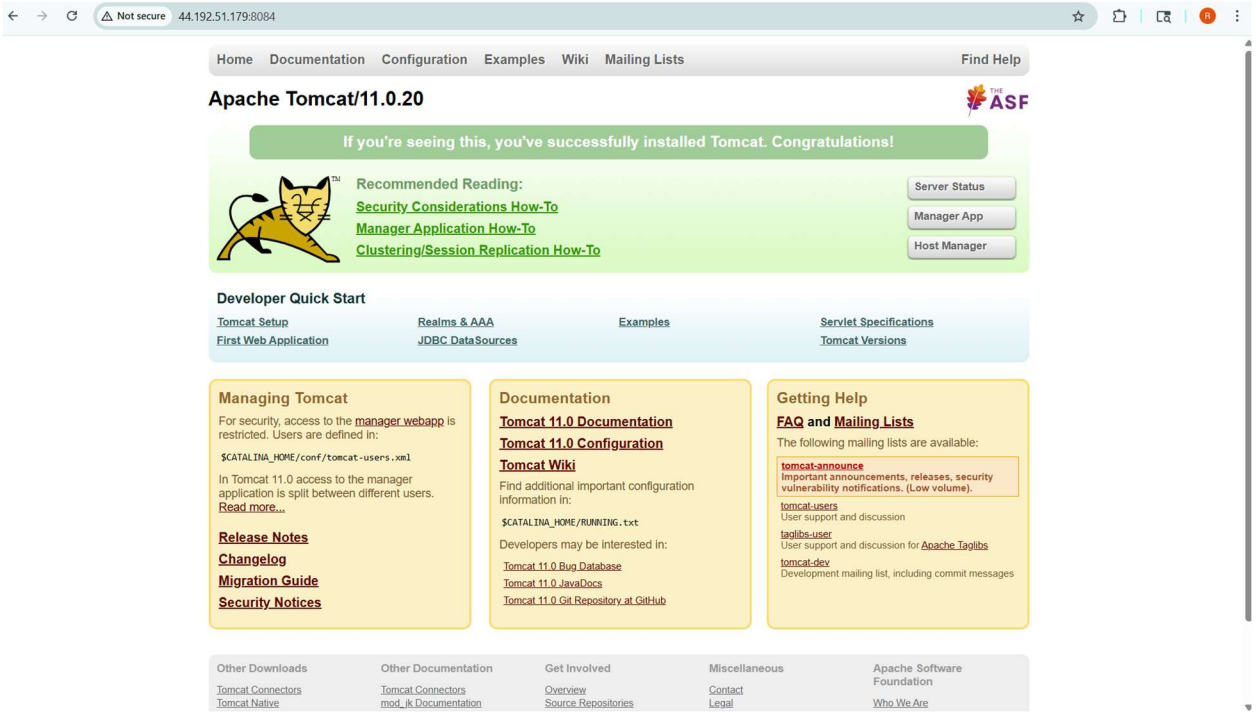
- Installed Apache Tomcat and successfully running

- 
```
[ec2-user@ip-172-31-67-252 ~]$ ps -ef | grep tomcat
root       39913       1  0 12:10 pts/3    00:00:12 /usr/bin/java -Djava.util.lo
gging.config.file=/opt/tomcat/conf/logging.properties -Djava.util.logging.manage
r=org.apache.juli.ClassLoaderLogManager -Djdk.tls.ephemeralDHKeySize=2048 -Dorg.
apache.catalina.security.SecurityListener.UMASK=0027 --add-opens=java.base/java.
lang=ALL-UNNAMED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens
=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.util=ALL-UNNAMED --add
-opens=java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.t
ransport=ALL-UNNAMED --enable-native-access=ALL-UNNAMED -classpath /opt/tomcat/b
in/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar -Dcatalina.base=/opt/tomcat -Dc
atalina.home=/opt/tomcat -Djava.io.tmpdir=/opt/tomcat/temp org.apache.catalina.s
tartup.Bootstrap start
ec2-user   42464   42223  0 13:00 pts/4    00:00:00 grep --color=auto tomcat
[ec2-user@ip-172-31-67-252 ~]$ sudo ss -tulnp | grep 8084
tcp   LISTEN 0      100                                  *:8084            *:*
    users:(("java",pid=39913,fd=43))

[ec2-user@ip-172-31-67-252 ~]$ |
```

- 

all services are properly installed, configured, and running on the same EC2 instance.
changed port numbers as needed to avoid conflicts